

Physics-Informed DMD (piDMD)

Aim: Learn structured linear models from data.

~~Recall~~ Given snapshot matrices

$$X = \begin{bmatrix} 1 & & 1 \\ x_1 & \dots & x_m \\ 1 & & 1 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 1 & & 1 \\ y_1 & \dots & y_m \\ 1 & & 1 \end{bmatrix},$$

$d \times m$ $d \times m$

where $y_j = T x_j \quad j=1, \dots, m$

DMD models minimize the objective:

$$\hat{T}_{\text{DMD}} = \underset{S \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \|Y - SX\|_F,$$

and $\hat{T}_{\text{DMD}} = YX^\dagger$ is low-rank when $m \ll n$.

piDMD constrains S to a matrix manifold:

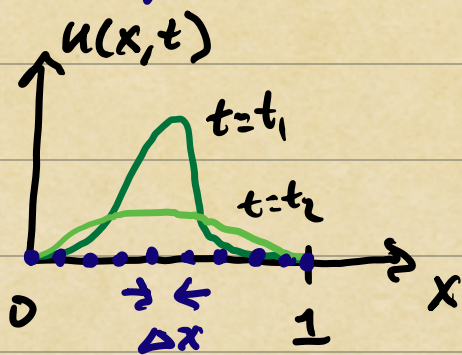
$$\hat{T}_{\text{pi}} = \underset{S \in M}{\operatorname{argmin}} \|Y - SX\|_F.$$

The manifold M encodes the physical properties of the dynamical system:

$$x_{k+1} = T x_k, \quad k=0, 1, 2, \dots$$

which constrain the entries/structure of T .

Example. 1D Diffusion



$$\begin{array}{ll} \text{PDE} & \partial_t u(x, t) = \partial_x^2 u(x, t) \\ \text{I.C.} & u(x, 0) = g(x) \\ \text{B.C.} & u(0, t) = u(1, t) = 0 \end{array}$$

Q: What is a "discrete" dynamical system associated w/the 1D Diffusion Eqn?

$$\frac{d}{dt} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_d \end{bmatrix} \approx \frac{1}{(\Delta x)^2} \begin{bmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_d \end{bmatrix}$$

A

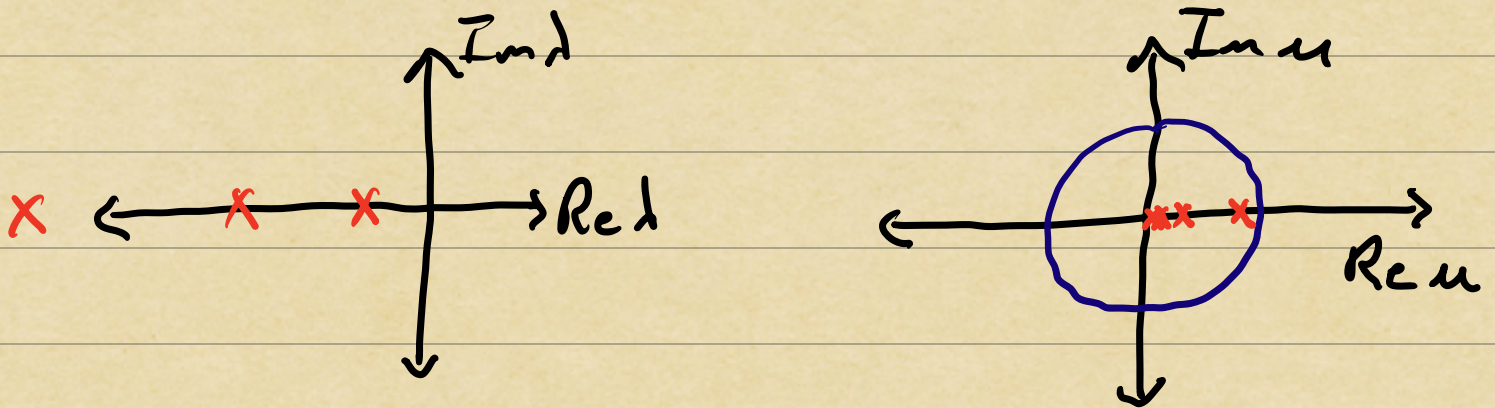
$$u_k = u(x_k, t)$$

$$\Rightarrow \text{Use } \partial_x^2 u|_{x_k} \approx \frac{1}{\Delta x^2} [u_{k+1} - 2u_k + u_{k-1}]$$

The discrete-time model for time step Δt :

$$\bar{T}_{\Delta t} = \exp(A\Delta t) = V \exp(\Lambda \Delta t) V^T.$$

The eigenvalues of A are real and negative:



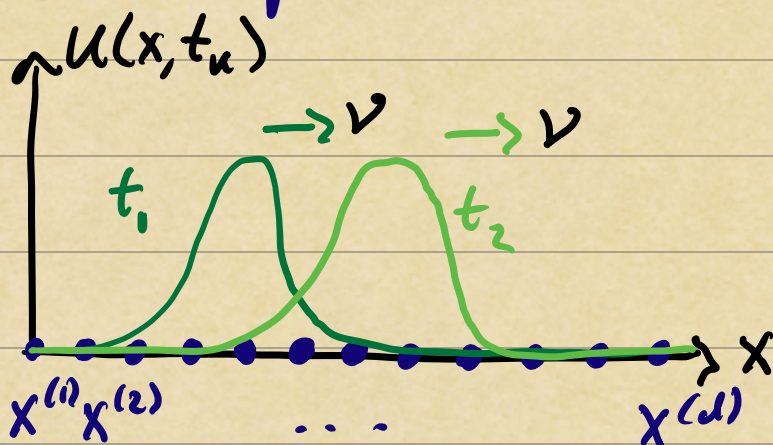
The eigenvalues of T are real and decay exponentially fast to zero.

Q: Do you expect DMD to find a good model for diffusion?

Q: What other physical properties might you want to "build" into DMD?

$\Rightarrow$ Self-adjoint A = real eigenvalues

Example. 1D Advection



$$\begin{aligned} \partial_t u &= v \partial_x u \\ u(x, 0) &= g(x) \\ u(0, t) &= u(1, t) \end{aligned}$$

Q: What is the "discrete" dynamical system associated w/ the 1D Advection Egn?

Two possible choices for the discrete model:

$$\frac{d}{dt} \begin{bmatrix} u^{(1)} \\ u^{(2)} \\ \vdots \\ u^{(d)} \end{bmatrix} \approx \frac{1}{2\Delta x} \begin{bmatrix} 0 & 1 & & -1 \\ -1 & 0 & 1 & \\ & \ddots & \ddots & \ddots \\ 1 & & -1 & 0 \end{bmatrix} \begin{bmatrix} u^{(1)} \\ u^{(2)} \\ \vdots \\ u^{(d)} \end{bmatrix}$$

OR

$$\frac{d}{dt} \begin{bmatrix} u^{(1)} \\ u^{(2)} \\ \vdots \\ u^{(d)} \end{bmatrix} \approx \frac{1}{\Delta x} \begin{bmatrix} 1 & & & -1 \\ -1 & 1 & & \\ & \ddots & \ddots & \ddots \\ & & -1 & 1 \end{bmatrix} \begin{bmatrix} u^{(1)} \\ u^{(2)} \\ \vdots \\ u^{(d)} \end{bmatrix}$$

$$\Rightarrow u_{k+1} \approx \exp(A\Delta t) u_k, \quad k=0, 1, 2, \dots$$

Q: Do you expect DMD to find a good model for advection?

Q: What physical properties might you build into p-DMD to improve solution?

$\Rightarrow$ Causal (Info flows only rightward)

$\Rightarrow$ Shift-invariant (Dynamics commute w/shifts)

$\Rightarrow$ Conservative (2-norm of soln. preserved)

$\Rightarrow$ Time-Periodic (Soln is periodic in $t \gg 0$)